PADHAKU
Mathematics · Class 10 Mathematics

CHAPTER 13

Statistics

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡 answers are samples: write your own.

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In-text questions and activities

13.2 ? **Tables 13.1 and 13.3 use the same data and the same formula, but the results (59.3 and 62) are different. Can you think why this is so, and which one is more accurate?**

✔ ANSWER

Table 13.1 uses the actual marks of each student, so 59.3 is the **exact mean**.

Table 13.3 uses grouped data: every student of a class is assumed to have the class mark (mid-point) as marks. This mid-point assumption changes the sum $\Sigma f_i x_i$, so 62 is only an **approximate mean**.

59.3 (ungrouped data) is more accurate.

Activity 1 **From Table 13.3 find the mean by taking each of x_i (17.5, 32.5, ...) as 'a'. What do you observe? Why is the mean the same each time?**

✔ ANSWER

Taking a = 17.5: $\Sigma f_i d_i = 30 \times (62 - 17.5) = 1335$, mean = 17.5 + 1335/30 = **62**.

Taking a = 32.5: $\Sigma f_i d_i = 885$, mean = 32.5 + 885/30 = **62**; similarly for 47.5, 62.5, 77.5 and 92.5.

Observation: the mean is **62 in every case**.

Why: $\bar{x} = a + \Sigma f_i d_i / \Sigma f_i$ and $d_i = x_i - a$, so $\Sigma f_i d_i = \Sigma f_i x_i - a \cdot \Sigma f_i$. Then $\bar{x} = a + \Sigma f_i x_i / \Sigma f_i - a = \Sigma f_i x_i / \Sigma f_i$. The 'a' cancels out, so the mean does not depend on the choice of a.

Activity 2 **Collect (1) Mathematics marks of your class, (2) daily maximum temperatures for 30 days, (3) heights of students; form grouped frequency tables and find the mean by a suitable method.**

💡 **SAMPLE ANSWER**

Model answer (do it with your own class data):

- Choose equal class intervals (e.g. marks 0–10, 10–20, ...; temperature 30–32 °C, 32–34 °C, ...; height 140–145 cm, 145–150 cm, ...) and count the tally in each class to get f_i.
- Find class marks x_i = (upper limit + lower limit)/2.
- If x_i and f_i are small, use the direct method $\bar{x} = \Sigma f_i x_i / \Sigma f_i$; if the numbers are large, use the assumed mean method $\bar{x} = a + \Sigma f_i d_i / \Sigma f_i$ or the step-deviation method $\bar{x} = a + (\Sigma f_i u_i / \Sigma f_i) \times h$.
- Example: heights 140–145 (3), 145–150 (8), 150–155 (12), 155–160 (5), 160–165 (2): $a = 152.5$, $h = 5$, $\Sigma f_i u_i = -6 + -8 + 0 + 5 + 4 = -5$, $\Sigma f_i = 30$, mean $= 152.5 + (-5/30) \times 5 =$ **151.67 cm**.

Activity 3 **For the same data as in Activity 2, find the mode. Compare it with the mean and interpret both.**

💡 SAMPLE ANSWER

Model answer (with the sample heights above):

- Modal class = 150–155 (highest frequency 12), $l = 150$, $h = 5$, $f_1 = 12$, $f_o = 8$, $f_2 = 5$.
- Mode = $150 + (12 - 8)/(24 - 8 - 5) \times 5 = 150 + 20/11$ = **151.82 cm**.
- Mean = 151.67 cm (from Activity 2).
- Interpretation: the largest number of students are about 151.8 cm tall, while the average height of a student is about 151.7 cm. Here the two values are nearly equal; in other data the mode may be less than or more than the mean.

13.4 (Why?) **For the marks of 100 students (20, 29, 28, 33, 42, 38, 43, 25 with frequencies 6, 28, 24, 15, 2, 4, 1, 20), why is the median 28?**

✔ ANSWER

n = 100 (even), so the median is the average of the 50th and 51st observations.

Arrange in ascending order with cumulative frequency: 20 → 6, 25 → 26, 28 → 50, 29 → 78, ...

The 27th to 50th observations are all 28, so the 50th observation is 28; the 51st to 78th are 29, so the 51st observation is 29.

Median = (28 + 29)/2 = **28.5**.

Exercise 13.1

1 **A survey collected the number of plants in 20 houses. Find the mean number of plants per house. Which method did you use, and why?**

Number of plants	0–2	2–4	4–6	6–8	8–10	10–12	12–14
Number of houses	1	2	1	5	6	2	3

✔ ANSWER

Class marks x_i = (lower + upper)/2.

Class	f_i	x_i	$f_i x_i$
0–2	1	1	1
2–4	2	3	6
4–6	1	5	5
6–8	5	7	35
8–10	6	9	54
10–12	2	11	22
12–14	3	13	39
Total	20		162

Mean $\bar{x} = \Sigma f_i x_i / \Sigma f_i = 162/20 =$ **8.1 plants per house**.

Method: the **direct method**, because the values of x_i and f_i are small, so the products $f_i x_i$ are easy to calculate.

2 **Daily wages of 50 workers of a factory. Find the mean daily wages by an appropriate method.**

Daily wages (in ₹)	500–520	520–540	540–560	560–580	580–600
Number of workers	12	14	8	6	10

✔ ANSWER

The values are large, so use the step-deviation method with $a = 550$, $h = 20$, $u_i = (x_i - 550)/20$.

Class	f_i	x_i	u_i	f_iu_i
500–520	12	510	−2	−24
520–540	14	530	−1	−14
540–560	8	550	0	0
560–580	6	570	1	6
580–600	10	590	2	20
Total	50			−12

$\bar{x} = a + (\Sigma f_iu_i/\Sigma f_i) \times h = 550 + (-12/50) \times 20 = 550 - 4.8 = 545.2$

Mean daily wage = ₹ 545.20

3 **The mean pocket allowance is ₹ 18. Find the missing frequency f.**

Daily pocket allowance (in ₹)	11–13	13–15	15–17	17–19	19–21	21–23	23–25
Number of children	7	6	9	13	f	5	4

✔ ANSWER

Class	f_i	x_i	f_ix_i
11–13	7	12	84
13–15	6	14	84
15–17	9	16	144
17–19	13	18	234
19–21	f	20	20f
21–23	5	22	110
23–25	4	24	96
Total	44 + f		752 + 20f

Mean = $\Sigma f_ix_i/\Sigma f_i \Rightarrow 18 = (752 + 20f)/(44 + f)$

$18(44 + f) = 752 + 20f \Rightarrow 792 + 18f = 752 + 20f$

$2f = 40 \Rightarrow$ **f = 20**

4 **Heartbeats per minute of 30 women. Find the mean heartbeats per minute, choosing a suitable method.**

Number of heartbeats per minute	65–68	68–71	71–74	74–77	77–80	80–83	83–86
Number of women	2	4	3	8	7	4	2

✔ ANSWER

Step-deviation method: a = 75.5, h = 3, $u_i = (x_i - 75.5)/3$.

Class	f_i	x_i	u_i	f_iu_i
65–68	2	66.5	−3	−6
68–71	4	69.5	−2	−8
71–74	3	72.5	−1	−3
74–77	8	75.5	0	0
77–80	7	78.5	1	7
80–83	4	81.5	2	8
83–86	2	84.5	3	6
Total	30			4

$\bar{x} = 75.5 + (4/30) \times 3 = 75.5 + 0.4 = 75.9$

Mean = 75.9 heartbeats per minute

5 **Distribution of mangoes according to the number of boxes. Find the mean number of mangoes in a packing box. Which method did you choose?**

Number of mangoes	50–52	53–55	56–58	59–61	62–64

Number of boxes	**15**	**110**	**135**	**115**	**25**

✔ ANSWER

The class marks are 51, 54, 57, 60, 63 (same for the continuous classes 49.5–52.5, 52.5–55.5, …; h = 3).

Step-deviation method with a = 57, h = 3:

Class	f_i	x_i	u_i	f_iu_i
50–52	15	51	−2	−30
53–55	110	54	−1	−110
56–58	135	57	0	0
59–61	115	60	1	115
62–64	25	63	2	50
Total	400			25

$\bar{x} = 57 + (25/400) \times 3 = 57 + 0.1875 = 57.1875$

Mean = 57.19 mangoes per box (approx.)

Method chosen: the **step-deviation method**, because f_i are large and the class marks have equal gaps (h = 3), which makes the calculation short.

6 Daily expenditure on food of 25 households. Find the mean daily expenditure by a suitable method.

Daily expenditure (in ₹)	100–150	150–200	200–250	250–300	300–350
Number of households	4	5	12	2	2

✔ ANSWER

Step-deviation method: a = 225, h = 50.

Class	f_i	x_i	u_i	f_iu_i
100–150	4	125	−2	−8
150–200	5	175	−1	−5
200–250	12	225	0	0
250–300	2	275	1	2
300–350	2	325	2	4
Total	25			−7

$\bar{x} = 225 + (-7/25) \times 50 = 225 - 14 = 211$

Mean daily expenditure on food = ₹ 211

7 Concentration of SO_2 (in ppm) in 30 localities. Find the mean concentration of SO_2 in the air.

Concentration of SO_2 (in ppm)	Frequency
0.00–0.04	4
0.04–0.08	9
0.08–0.12	9
0.12–0.16	2
0.16–0.20	4

0.20–0.24	2

✔ ANSWER

Direct method:

Class	f_i	x_i	f_ix_i
0.00–0.04	4	0.02	0.08
0.04–0.08	9	0.06	0.54
0.08–0.12	9	0.10	0.90
0.12–0.16	2	0.14	0.28
0.16–0.20	4	0.18	0.72
0.20–0.24	2	0.22	0.44
Total	30		2.96

$\bar{x}$ = 2.96/30 = 0.0987 (approx.)

Mean concentration of SO_2 = 0.099 ppm (approx.)

8 **Absentee record of 40 students for the whole term. Find the mean number of days a student was absent.**

Number of days	0–6	6–10	10–14	14–20	20–28	28–38	38–40
Number of students	11	10	7	4	4	3	1

✔ ANSWER

Class sizes are unequal, so use the assumed mean method with a = 17 ($d_i = x_i - 17$).

Class	f_i	x_i	d_i	f_id_i
0–6	11	3	−14	−154

6–10	10	8	−9	−90
10–14	7	12	−5	−35
14–20	4	17	0	0
20–28	4	24	7	28
28–38	3	33	16	48
38–40	1	39	22	22
Total	40			−181

$\bar{x} = a + \Sigma f_i d_i / \Sigma f_i = 17 + (-181/40) = 17 - 4.525 = 12.475$

Mean = 12.48 days (approx.)

9 **Literacy rate (in %) of 35 cities. Find the mean literacy rate.**

Literacy rate (in %)	45–55	55–65	65–75	75–85	85–95
Number of cities	3	10	11	8	3

✔ ANSWER

Step-deviation method: a = 70, h = 10.

Class	f_i	x_i	u_i	$f_i u_i$
45–55	3	50	−2	−6
55–65	10	60	−1	−10
65–75	11	70	0	0
75–85	8	80	1	8
85–95	3	90	2	6
Total	35			−2

$\bar{x} = 70 + (-2/35) \times 10 = 70 - 0.571 = 69.43$

Mean literacy rate = 69.43 % (approx.)

Exercise 13.2

1 **Ages of patients admitted in a hospital during a year. Find the mode and the mean. Compare and interpret the two measures.**

Age (in years)	5–15	15–25	25–35	35–45	45–55	55–65
Number of patients	6	11	21	23	14	5

✔ ANSWER

Mode: maximum frequency 23 → modal class 35–45.

$l = 35, h = 10, f_1 = 23, f_0 = 21, f_2 = 14$

Mode $= l + (f_1 - f_0)/(2f_1 - f_0 - f_2) \times h = 35 + (23 - 21)/(46 - 21 - 14) \times 10 = 35 + 2/11 \times 10 = 35 + 1.82 =$ **36.8 years** (approx.)

Mean (step-deviation, a = 30, h = 10):

Class	f_i	x_i	u_i	f_iu_i
5–15	6	10	−2	−12
15–25	11	20	−1	−11
25–35	21	30	0	0
35–45	23	40	1	23
45–55	14	50	2	28
55–65	5	60	3	15
Total	80			43

$\bar{x} = 30 + (43/80) \times 10 = 30 + 5.375 =$ **35.37 years** (approx.)

Interpretation: the maximum number of patients admitted are of age about 36.8 years, while on an average a patient admitted is about 35.37 years old.

2 **Observed lifetimes (in hours) of 225 electrical components. Determine the modal lifetime.**

Lifetimes (in hours)	0–20	20–40	40–60	60–80	80–100	100–120
Frequency	10	35	52	61	38	29

✔ ANSWER

Maximum frequency 61 → modal class 60–80.

$l = 60$, $h = 20$, $f_1 = 61$, $f_o = 52$, $f_2 = 38$

Mode = $60 + (61 - 52)/(122 - 52 - 38) \times 20 = 60 + 9/32 \times 20 = 60 + 5.625$

Modal lifetime = 65.625 hours

3 **Total monthly household expenditure of 200 families. Find the modal monthly expenditure and the mean monthly expenditure.**

Expenditure (in ₹)	Number of families
1000–1500	24
1500–2000	40
2000–2500	33
2500–3000	28
3000–3500	30
3500–4000	22
4000–4500	16

4500–5000	7

✔ ANSWER

Mode: maximum frequency 40 → modal class 1500–2000.

$l = 1500$, $h = 500$, $f_1 = 40$, $f_o = 24$, $f_2 = 33$

Mode = $1500 + (40 - 24)/(80 - 24 - 33) \times 500 = 1500 + 16/23 \times 500 = 1500 + 347.83$

Modal monthly expenditure = ₹ 1847.83 (approx.)

Mean (step-deviation, $a = 2750$, $h = 500$):

Class	f_i	x_i	u_i	f_iu_i
1000–1500	24	1250	−3	−72
1500–2000	40	1750	−2	−80
2000–2500	33	2250	−1	−33
2500–3000	28	2750	0	0
3000–3500	30	3250	1	30
3500–4000	22	3750	2	44
4000–4500	16	4250	3	48
4500–5000	7	4750	4	28
Total	200			−35

$\bar{x} = 2750 + (-35/200) \times 500 = 2750 - 87.5 = 2662.5$

Mean monthly expenditure = ₹ 2662.50

4 **State-wise teacher-student ratio in higher secondary schools of India. Find the mode and mean. Interpret the two measures.**

Number of students per teacher	Number of states / U.T.
15–20	3
20–25	8
25–30	9
30–35	10
35–40	3
40–45	0
45–50	0
50–55	2

✔ ANSWER

Mode: maximum frequency 10 → modal class 30–35.

$l = 30$, $h = 5$, $f_1 = 10$, $f_0 = 9$, $f_2 = 3$

Mode $= 30 + (10 - 9)/(20 - 9 - 3) \times 5 = 30 + 1/8 \times 5 = 30 + 0.625 =$ **30.6** (approx.)

Mean (step-deviation, $a = 32.5$, $h = 5$):

Class	f_i	x_i	u_i	f_iu_i
15–20	3	17.5	−3	−9
20–25	8	22.5	−2	−16
25–30	9	27.5	−1	−9
30–35	10	32.5	0	0
35–40	3	37.5	1	3
40–45	0	42.5	2	0
45–50	0	47.5	3	0
50–55	2	52.5	4	8
Total	35			−23

$\bar{x}$ = 32.5 + (−23/35) × 5 = 32.5 − 3.29 = **29.2** (approx.)

Interpretation: most states/U.T. have a student-teacher ratio of about 30.6, while on an average this ratio is about 29.2.

5 **Runs scored by some top batsmen in one-day international matches. Find the mode of the data.**

Runs scored	Number of batsmen
3000–4000	4
4000–5000	18
5000–6000	9
6000–7000	7
7000–8000	6
8000–9000	3
9000–10000	1
10000–11000	1

✔ ANSWER

Maximum frequency 18 → modal class 4000–5000.

l = 4000, h = 1000, f_1 = 18, f_0 = 4, f_2 = 9

Mode = 4000 + (18 − 4)/(36 − 4 − 9) × 1000 = 4000 + 14/23 × 1000 = 4000 + 608.7

Mode = 4608.7 runs (approx.)

6 **Number of cars passing a spot in 100 periods of 3 minutes each. Find the mode of the data.**

Number of cars	0–10	10–20	20–30	30–40	40–50	50–60	60–70	70–80
Frequency	7	14	13	12	20	11	15	8

✔ ANSWER

Maximum frequency 20 → modal class 40–50.

$l = 40$, $h = 10$, $f_1 = 20$, $f_o = 12$, $f_2 = 11$

Mode = $40 + (20 - 12)/(40 - 12 - 11) \times 10 = 40 + 8/17 \times 10 = 40 + 4.7$

Mode = 44.7 cars (approx.)

Exercise 13.3

1 **Monthly consumption of electricity of 68 consumers. Find the median, mean and mode of the data and compare them.**

Monthly consumption (in units)	Number of consumers
65–85	4
85–105	5
105–125	13
125–145	20
145–165	14
165–185	8
185–205	4

✔ ANSWER

Median: $n = 68$, $n/2 = 34$.

Class	f_i	cf
65–85	4	4

85–105	5	9
105–125	13	22
125–145	20	42
145–165	14	56
165–185	8	64
185–205	4	68

cf just greater than 34 is 42 → median class 125–145: l = 125, h = 20, f = 20, cf = 22.

Median = l + (n/2 − cf)/f × h = 125 + (34 − 22)/20 × 20 = 125 + 12 = **137 units**

Mean (step-deviation, a = 135, h = 20): x_i = 75, 95, 115, 135, 155, 175, 195; u_i = −3, −2, −1, 0, 1, 2, 3

$\Sigma f_i u_i$ = −12 − 10 − 13 + 0 + 14 + 16 + 12 = 7

$\bar{x}$ = 135 + (7/68) × 20 = 135 + 2.06 = **137.05 units** (approx.)

Mode: modal class 125–145 (f_1 = 20), l = 125, h = 20, f_o = 13, f_2 = 14

Mode = 125 + (20 − 13)/(40 − 13 − 14) × 20 = 125 + 7/13 × 20 = 125 + 10.77 = **135.77 units** (approx.)

Comparison: median 137, mean 137.05 and mode 135.77 are approximately equal; the three measures are nearly the same for this data.

2 **If the median of the distribution is 28.5, find the values of x and y.**

Class interval	Frequency
0–10	5

10–20	x
20–30	20
30–40	15
40–50	y
50–60	5
Total	60

✔ ANSWER

Total: 5 + x + 20 + 15 + y + 5 = 60 ⇒ x + y = 15 … (1)

Class	f_i	cf
0–10	5	5
10–20	x	5 + x
20–30	20	25 + x
30–40	15	40 + x
40–50	y	40 + x + y
50–60	5	45 + x + y

Median 28.5 lies in class 20–30 → l = 20, h = 10, f = 20, cf = 5 + x, n/2 = 30.

28.5 = 20 + (30 − 5 − x)/20 × 10 ⇒ 8.5 = (25 − x)/2 ⇒ 25 − x = 17 ⇒ **x = 8**

From (1): y = 15 − 8 ⇒ **y = 7**

3 **Ages of 100 policy holders (policies are given only to persons aged 18 years or more but less than 60). Calculate the median age.**

Age (in years)	Number of policy holders
Below 20	2
Below 25	6

Below 30	24
Below 35	45
Below 40	78
Below 45	89
Below 50	92
Below 55	98
Below 60	100

✔ ANSWER

The table gives cumulative frequencies (less than type). Convert to class intervals; the first class starts at 18.

Class	f_i	cf
18–20	2	2
20–25	4	6
25–30	18	24
30–35	21	45
35–40	33	78
40–45	11	89
45–50	3	92
50–55	6	98
55–60	2	100

n = 100, n/2 = 50; cf just greater than 50 is 78 → median class 35–40: l = 35, h = 5, f = 33, cf = 45.

Median = 35 + (50 − 45)/33 × 5 = 35 + 25/33 = 35 + 0.76

Median age = 35.76 years (approx.)

4 **Lengths of 40 leaves measured to the nearest millimetre. Find the median length. (Hint: convert to continuous classes 117.5–126.5, 126.5–135.5, ...)**

Length (in mm)	Number of leaves
118–126	3
127–135	5
136–144	9
145–153	12
154–162	5
163–171	4
172–180	2

✔ ANSWER

Continuous classes (subtract 0.5 from lower limits, add 0.5 to upper limits):

Class	f_i	cf
117.5–126.5	3	3
126.5–135.5	5	8
135.5–144.5	9	17
144.5–153.5	12	29
153.5–162.5	5	34
162.5–171.5	4	38
171.5–180.5	2	40

n = 40, n/2 = 20; cf just greater than 20 is 29 → median class 144.5–153.5: l = 144.5, h = 9, f = 12, cf = 17.

Median = 144.5 + (20 − 17)/12 × 9 = 144.5 + 27/12 = 144.5 + 2.25

Median length = 146.75 mm

5 **Life time of 400 neon lamps. Find the median life time of a lamp.**

Life time (in hours)	Number of lamps
1500–2000	14
2000–2500	56
2500–3000	60
3000–3500	86
3500–4000	74
4000–4500	62
4500–5000	48

✔ ANSWER

Class	f_i	cf
1500–2000	14	14
2000–2500	56	70
2500–3000	60	130
3000–3500	86	216
3500–4000	74	290
4000–4500	62	352
4500–5000	48	400

n = 400, n/2 = 200; cf just greater than 200 is 216 → median class 3000–3500: l = 3000, h = 500, f = 86, cf = 130.

Median = 3000 + (200 − 130)/86 × 500 = 3000 + 35000/86 = 3000 + 406.98

Median life time = 3406.98 hours (approx.)

6 **Number of letters in 100 surnames. Determine the median number of letters. Find the mean number of letters. Also find the modal size of the surnames.**

Number of letters	1–4	4–7	7–10	10–13	13–16	16–19
Number of surnames	6	30	40	16	4	4

✔ ANSWER

Median:

Class	f_i	x_i	cf
1–4	6	2.5	6
4–7	30	5.5	36
7–10	40	8.5	76
10–13	16	11.5	92
13–16	4	14.5	96
16–19	4	17.5	100

n = 100, n/2 = 50; cf just greater than 50 is 76 → median class 7–10: l = 7, h = 3, f = 40, cf = 36.

Median = 7 + (50 − 36)/40 × 3 = 7 + 42/40 = 7 + 1.05 = **8.05 letters**

Mean (step-deviation, a = 8.5, h = 3): u_i = −2, −1, 0, 1, 2, 3

$\Sigma f_i u_i$ = −12 − 30 + 0 + 16 + 8 + 12 = −6

$\bar{x}$ = 8.5 + (−6/100) × 3 = 8.5 − 0.18 = **8.32 letters**

Mode: modal class 7–10 (f_1 = 40), l = 7, h = 3, f_o = 30, f_2 = 16

Mode = 7 + (40 − 30)/(80 − 30 − 16) × 3 = 7 + 10/34 × 3 = 7 + 0.88 = **7.88 letters** (approx.)

7 **Weights of 30 students of a class. Find the median weight of the students.**

Weight (in kg)	40–45	45–50	50–55	55–60	60–65	65–70	70–75
Number of students	2	3	8	6	6	3	2

✔ ANSWER

Class	f_i	cf
40–45	2	2
45–50	3	5
50–55	8	13
55–60	6	19
60–65	6	25
65–70	3	28
70–75	2	30

n = 30, n/2 = 15; cf just greater than 15 is 19 → median class 55–60: l = 55, h = 5, f = 6, cf = 13.

Median = 55 + (15 − 13)/6 × 5 = 55 + 10/6 = 55 + 1.67

Median weight = 56.67 kg (approx.)

🎉 **Chapter 13 done! Revise the answers once more tomorrow.**